BHARATHWAJ NEDOUMARAN

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EDUCATION

PURDUE UNIVERSITY | Master's in Computer Science | GPA: 4.0 | INDIANA, USA **AUGUST 2025 – MAY 2027**

ANNA UNIVERSITY | Bachelor's in Computer Science | GPA: 3.5 | TAMILNADU, INDIA **AUGUST 2019 – MAY 2023**

PROFESSIONAL EXPERIENCE

ORACLE | APPLICATION DEVELOPER II **SEPTEMBER 2023 – JULY 2025**

- **Orchestrated Cross-Platform Release Validation:** Directed installation, security, and performance testing protocols for the **Oracle JD Edwards EnterpriseOne** applications across **Windows, Linux, and Unix** environments to ensure zero-defect production deployments.
- **Architected Full-Stack Release Platform:** Developed a comprehensive Release Engineering Dashboard **(Oracle APEX, JavaScript, SQL)** acting as the centralized frontend to manage all VMs, servers, test results, and bug databases across **30+ concurrent enterprise projects**.
- **Engineered Backend Automation Tooling:** Built and securely deployed the supporting **Python Flask REST APIs** via **Nginx** on Linux VMs to process dashboard-driven commands, automating manual infrastructure maintenance, database schema validations, and cross-platform service restarts.
- **Accelerated Continuous Delivery:** Streamlined the **CI/CD pipeline** by enabling one-click automated test triggers directly from the dashboard UI, reducing manual deployment validation effort by **25%** and accelerating overall delivery lifecycles by **40%**.

SKILLS

- **Languages:** Java, Python, GoLang, HTML, CSS, JavaScript, SQL, PL/SQL
- **Web & Frameworks:** Spring, Flask, Django, Beego, React.js, Node.js, FastAPI, Microservices
- **Database:** MySQL, PostgreSQL, Redis, Oracle Database, IBM DB2, MongoDB
- **Cloud & DevOps:** AWS, OCI, Docker, Kubernetes, Henkins, Ansible, CI/CI Pipelines
- **Testing & Security:** Playwright, OAuth 2.0, SAML (SSO), JWT
- **AI & ML:** Scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, PyTorch, OpenCV
- **Developer Tools:** Git/GitHub/Bitbucket, Jira, Kanban, VS Code, Eclipse, PyCharm, Postman, Figma
- **Concepts:** OOPS, Data Structures, REST API, MVC Architecture, Agile, Unit Testing, Distributed Systems
- **Soft Skills:** Communication, Problem Solving, Leadership, Organized, Time-Management

PROJECTS

ENTERPRISE TELEHEALTH & INVENTORY PLATFORM | Java, React Node.js, PostgreSQL, Docker, Git **Spring 2026**

- **Architected Microservices Backend:** Developed a robust healthcare REST API leveraging **Java 17 and Spring Boot**, supporting core platform features including secure user authentication, medical inventory management, and telehealth scheduling.
- **Engineered-High Performance UI:** Built a responsive, scalable frontend architecture utilizing **React, Node.js, and Vite**, styled with **Tailwind CSS** to ensure a seamless, high-speed user experience across all devices.
- **Containerized System Architecture:** Orchestrated the entire multi-tier application stack (Backend API, Frontend UI, and **PostgreSQL** database) using **Docker**, guaranteeing consistent environments and accelerating deployment times.
- **Administered Distributed Version Control:** Managed isolated microservice repositories within **Bitbucket**, utilizing **Git** for strict version control and seamless integration between frontend and backend workflows.

ADVERSARIAL ATTACK DETECTION SYSTEM | Python, TensorFlow, CNN, Autoencoders, FGSM **Fall 2025**

- **Architected ML Defense Pipeline:** Engineered a hybrid **CNN-Autoencoder** architecture utilizing **Python and TensorFlow** to process 70,000 image samples and actively detect Fast Gradient Sign Method (FGSM) security vulnerabilities.
- **Engineered High-Precision Classifier:** Developed a Binary CNN detection model to reliably distinguish clean datasets from maliciously perturbed images, achieving a perfect **1.0 ROC-AUC** and a **99.8% F1-score**.
- **Optimized Model Resilience:** Executed robust adversarial training protocols by generating and deploying **120,000 augmented data samples**, successfully boosting model classification accuracy on attacked images from a baseline of 2.8% to 98.9%.